

$$A = \begin{bmatrix} 9 & 15 \\ 15 & 50 \end{bmatrix}$$

Acc. to Cholesky decomposition,

$$A = LL^T \quad (L = \text{Lower triangular})$$

$$A = \begin{bmatrix} l_{11} & 0 \\ l_{21} & l_{22} \end{bmatrix} \begin{bmatrix} l_{11} & l_{21} \\ 0 & l_{22} \end{bmatrix} \quad \left(L = \begin{bmatrix} l_{11} & 0 \\ l_{21} & l_{22} \end{bmatrix} \right)$$

$$\rightarrow \begin{bmatrix} l_{11}^2 & l_{11}l_{21} \\ l_{11}l_{21} & l_{21}^2 + l_{22}^2 \end{bmatrix} = \begin{bmatrix} 9 & 15 \\ 15 & 50 \end{bmatrix}$$

$$l_{11}^2 = 9 \rightarrow \underline{l_{11} = 3}$$

$$l_{11}l_{21} = 15 \rightarrow l_{21} = 15/3 = 5 \rightarrow \underline{l_{21} = 5}$$

$$l_{21}^2 + l_{22}^2 = 50$$

$$25 + l_{22}^2 = 50$$

$$\rightarrow \boxed{l_{22} = 5}$$